Unit -III

Assignment -1

Q.1] Explain MVC Architecture with a suitable diagram.

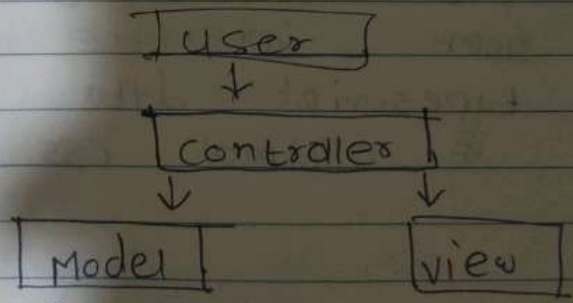
→

MVC (model-view-controuer) is a sentence architectural pattern that divides an application into there interconnected component: model, view & controller it makes the application easier to develop, test & maintain.

Component OF MVC;

1. model : manage application data & buiseness logic it interact & display database

Diagram



Advantages:

- seperate data, user interface & logic
- Makes code accurate & easier to maintain
- support parallel develop & testing

Q.2] P P Explain Angular component & pipes with suitable example.

→

1. Angular ~~com~~ Component

An Angular component is a including block angular application, it content a part of the user interface & consist of a typescript data & optimal css styles. an Html template

Example

```
import { Component } From
'@angular/core';

@Component({
    select : 'app-root';
    template : '<h1> Hello dynamic </h1>'
    })

export class App component {
    }
```

explanation:

- @ component define the component
- select specify the Html tag used to display

- template define the Html content.

- ~~App com~~me
- App component contain the component logic

2. dynamic pipes

pipes are used ~~tem~~ & Format data For display, in an HTML ~~to~~ template ~~to~~ without changing the orignal data

example:

```
export class AppComponent
{
    Name = "Atharv";
    today = new Date();
}
```

HTML

```
<P> {{ name | uppercase }} </P>
<P> {{ today | date: "dd/MM/yyyy" }} </P>
```

Output:

- ~~sidd~~ - Atharv
- current date is is dd/MM/yyyy Format.

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Q.3] Explain angular directive & Data Blindly with example

→

1. Angular Directives:

Directive are special instructions in Angular apperance, structure, or behaviours of HTML element

Type of directives

1. Component Directive

Directive with a template that define a part of the user interface.

2. structure Directive-
Add or remove HTML elements from the DOM

3. Attribute Directive:
Change the apperance or behaviour of existing element

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<~~Hee~~

example

<p *ngZP="" inloggen In"> welcome user]</P>

<li * ngFor ="let item of item">

2. Angular Data Binding.

Data binding connect the Component data with the HTML tt template ,allowing information to between ~~th~~ then.

- types of Data Binding

	Type	Syntax	purpose
①	interpolation	{{name}}	Display component data
②	property binding	[value]="name"	set an HTML property.

③ ~~Ex~~ Event binding (click)="show()" Handle user event

④ two-way binding [(ngModel)]="name" Sync data in direction

example

```
<h2> {{name}} </h2>

<input [value]="name" >

<button (click)="show()" > click the
                          </button>

<input [(ngModel)]="name">
```

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Assignment / Case Study No __ (Unit No 03 & 04)

Semester: I/ II/ III/ IV / V / VI / VII / VIII – A.Y.-20...- 20...

Name of the Student: .....

Year: FE/SE/TE/BE Div.: .....Roll No.: ..... Date.....

Supervisor's Name: ..... Signature: .....

Ass / CS	01	02	03	04	05	06	Total Marks	Sign. of Examiner
Marks Obtained							/	
COs Mapped								

Assignment - No - 2

Q1] Explain Express.js Routing & w & middlewa.. with a suitable program

→

Exprees .js Routing;

Routing in express .js define how a server respond when request for different URL & HTTP method such as GET, POST, PUT & DELETE

Example:

```
Count express = require ('express');
Const app = express()

app.get ('/', (req, res) =>{
    req. send ('welcome, Home');

});
```

```
app.get('/about', (req, res) => {
    res.send('about is page');
});

app.listen(3000);
```

Explanation

- express() create an express application
- app.get() define route for get request
- req repeat the request, & represent the response
- app.listen (3000) the server on Post 3000

Q.2] What is NOSQL ? Explain mongoDB Feature & component SQL with NOSQL

→ NoSQL.

NO SQL (not only SQL) is a type of database used to store & manage data without relying exclusively.

on the traditional taste structure Flexible data model & is useful For large-scale, application

Feature of MongoDB

* MongoDB is a document oriented NOSQL database that stores data in .JSON documents, a binary representation of JSON-like that

1. Document-oriented: stores data as document containing feild & values.

2. * Flexible scema- ~~sto~~ Documents in a collection & have different feild.

3. scalablity: support Horizontal scaling though sharing

4. High availblity: support replica sets for redundaring. ~~of fo~~ ~~facet~~.

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5. Indexing : improves the speed of data retrival

example mongo DB document

```
</> Javascript
{
    name : " Atharv",
    age : 64
    Course : "IT"
}
```

SQL Database	NoSQL Database
① stores data mainly is database	① store data in document key value pair, graph etc
② usually has a predefined schema	② often support a flexible schema
③ use SQL for queries	③ use database specific query method

④ example: MySQL, PosgeSQL, MicrosoftSQL, MariaDB	④ example: MongoDB, ~~Cassandra~~ Cassandra

Q.3] Explain crud operation in Node.js & MongoDB with suitable examples?

→ CRUD stand for create, read, update & delete, there are the four basic operation used to manage data in a database

~~First install~~

First, install the driver)

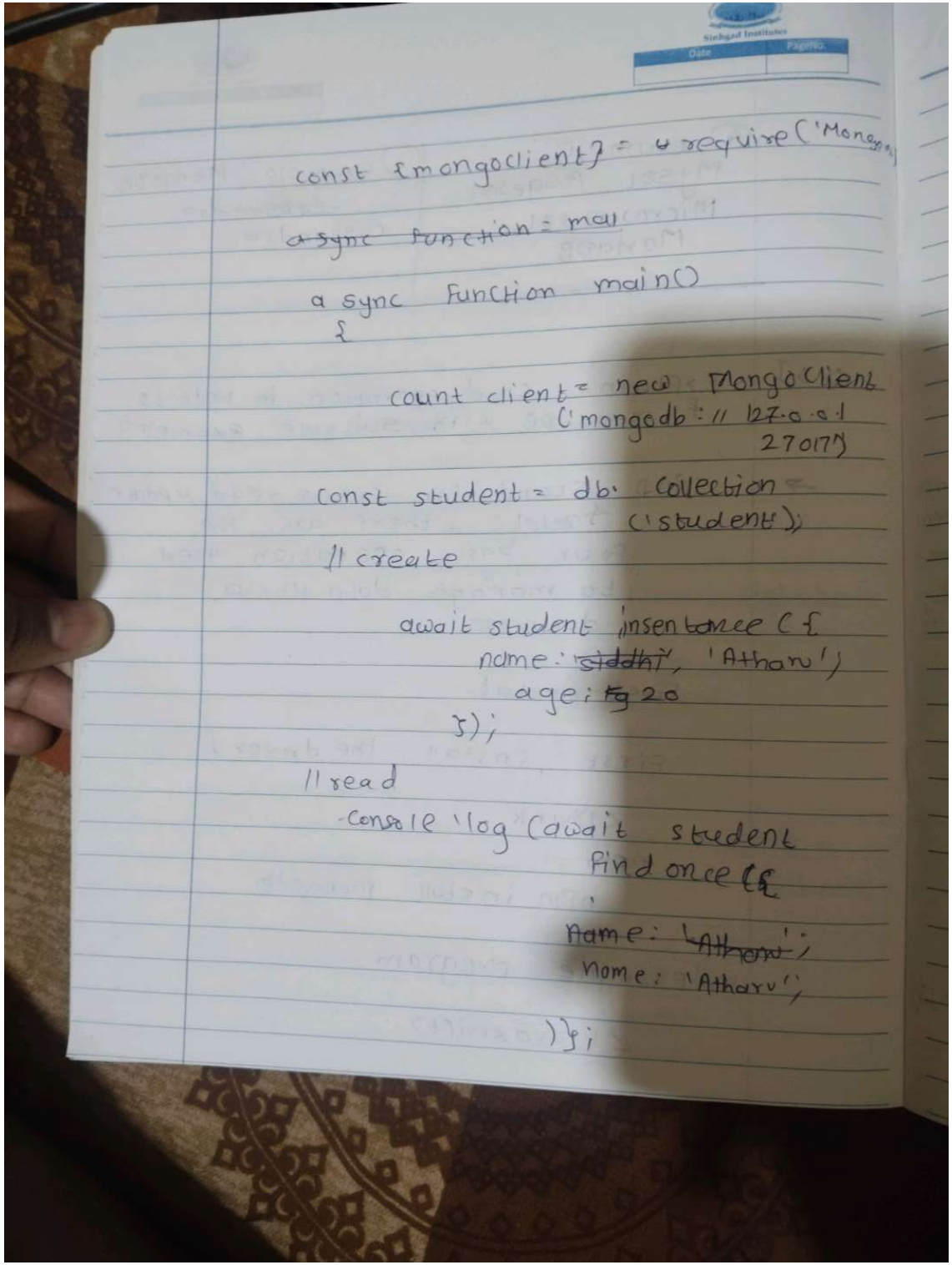
<1>bank

~~npm,~~

npm install mongodb

Example program

<Javascript>



```
const {mongoclient} = require('Mongo...

async function = mai

async function main()
{

    count client = new MongoClient
                ('mongodb://127.0.0.1
                27017')

const student = db.collection
                ('student');

// create

    await student.insertOne({
        name: 'siddhi', 'Atharv',
        age: 20
    });

// read
    console.log(await student
                findOne({

        name: 'Atharv';
        name: 'Atharv';

    }));
```

```
//update

await Student.updateOne(
    {name :'siddhi'}
    {name: 'Atharv'}
      {$set : {age:20}}

);

//Delete
    await student: deleteOne({
        name :'Atharv';

    });
        await client.user();

}
    main(); catch (console,error);
```

Q.4] Explain the Node.js five system & Demonstrate file operation

→ Nod.js file system module

due file system (fs) module is Node.js provides function to ~~perform~~ perform operation.

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it can be included using:

```
</>Javascript
const fs = require ('fs');
```

Common five operation

operation	method	purpose
write	fs.writeFileSync()	create or overwrite file
read	fs.read	Read file content
append	fs.append	Add content to a file
delete	fs.delete	Delete a file
rename	fs.rename	Rename a file

Example program

```
</> Javascript
const fs = require('fs');

// create & write to a file

fs.writeFileSync ('demo.txt'
,Hello Node.js');
```

// read the file

let data = fs.readFileSync('demo.txt', 'utf8');

console.log(data);

// Append data

fs.appendFileSync('demo.txt', '\n welcome!');

// Read update content

data = fs.readFileSync('demo.txt', 'utf8');

console.log(data)

// Rename the file

fs.renameSync('demo.txt', 'newdemo.txt');

// Delete the file

fs.unlinkSync('newdemo.txt');

Output:

Hello Node.js
Hello Node.js
welcome!

Conclusion: the Node.js file system module is useful for application that need to read, write, write, update, rename or delete files.